MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

SOCIALIST REPUBLIC OF VIETNAM
Independence - Freedom - Happiness

COURSE OUTLINE DETAILS

1. Course: Graduation thesis - Thesis for Software Engineering

- **Course number:** CT553H
- **Credits:** 10 credits
- **Hours:** 450 hours

2. Management unit:

- **Department:** Department of Software Engineering
- **Faculty:** Faculty of Information Technology and Communication

3. Conditions

- **Conditions precedent:** ≥120 credits
- **Corequisites:**

4. Course objectives:

Objectives	Description	ELOs
4.1	Identify a research topic in the field of software engineering.	2.1.3a, 2.1.3b, 2.2.1a
4.2	Apply knowledge of design principles, development processes, mathematical theories to build model, design and implement the correct solutions.	2.1.1b, 2.1.2, 2.1.3
4.3	Use software engineering tools to real-world projects.	2.2.1a, 2.2.1c
4.4	Present problems, solutions, and results via verbal and writing.	2.2.1b, 2.2.2a,
4. 5	Demonstrate professionalism in work and self-study to improve professional.	2.3b

5. Course learning outcomes:

CLOs	Descriptions	Objectives	ELOs
	Knowledge		
CLO 1	Apply knowledge of how to identify a research topic in the field of software	4.1	2.1.3a, 2.1.3b,

	engineering and to choose suitable research methods.		2.2.1a
CLO 2	Apply knowledge of mathematical theory, software engineering to solve a problem.	4.2	2.2.1a, 2.2.1c
	Skill		
CLO 3	Make and perform the plan to complete the thesis within the time limitation.	4.4	2.2.1a, 2.2.1c
CLO 4	Use tools to solve a problem.	4.3	2.2.1b
CLO 5	Apply and integrate existing knowledge with new knowledge to build solutions to problem.	4.2, 4.3	2.1.1b, 2.1.2, 2.1.3, 2.2.1a, 2.2.1c
CLO 6	Show skills in presenting the results via verbal and writing.	4.4	2.2.1b, 2.2.2
	Attitude/Autonomy/Responsibility		
CLO 7	Show the responsibility, the ability to self-study and the creativity in work.	4.5	2.3
CLO 8	Demonstrate the professional in work and in behavior when working with others.	4.5	2.3

6. Brief description of the course:

The thesis help students prove their capabilities for solving a complex problem in one semester. Students have to describe: the problem, the history of problem solving, concepts used by the thesis, solutions, test results, conclusions and future work in a report.

A Council will assess the result of students based on the report, the program and the presentation.

7. Course structure:

7.1. Theory

Students self-read the necessary references recommended by instructors.

No	Content	Hours	CLOs
1	Documentation related to software engineering such as: requirement analysis, design, quality assurance and testing, project management, and software maintenance.		CO1-CO2, CO5
2	Documentation related to the fundamental knowledge such as: data structures, algorithms, databases, programming languages.		CO1-CO2, CO5
3	Documentation of application field such as: mobile device software, business software, agent-oriented		CO1-CO2, CO5

	software, embedded software.		

7.2. Practice

Students follow the following tasks to achieve the objectives of the thesis.

No	Content	Hours	CLOs
1	Identify the requirements of problem		CO1-CO2
2	Make the plan		CO3
3	Study the history of problem solving.		CO1-CO3
4	Study theories and practices related to problem.		CO1-CO6
5	Propose solutions		CO2, CO4-CO5
6	Implement solution		CO3-CO5, CO7-CO8
7	Evaluate results		All COs
8	Write a report		CO3, CO6
9	Present results		All COs

7. Teaching methods:

- Theory: Students will read the documents introduced by the instruction set.
- Practice: Students will perform the necessary tasks to achieve the goal of the project; periodically report the results to instructors.

9. Duties of student:

Students have to do the following duties:

- Actively contact the instructors to get the acceptance.
- Have the responsibility to meet the instructors according to the agreed working schedule to report the completed tasks and ask for comments on the next tasks.
- Ensure work progress according to plan.
- Present the results in front of instructors.
- Submit the report.
- Present the results in front of Council.
- Complete and submit the product (the report and CD-ROM) to CICT.

10. Assessment of course learning outcomes:

10.1. Assessment

The score is evaluated mainly based on:

- Applicability of software engineering (project management, requirements analysis, design, quality assurance, testing, etc.).

- Applicability of mathematical theory, science, and computational skills to problem solving.
- Communicate effectively via verbal and writing.
- Present the results clearly and understandably.
- Working attitude.
- Creativity and contribution.

All of the above requirements are reflected in the performed tasks, the report, the program, the presentation:

Specific scores are described in the graduation thesis assessment rubric of the Software Engineering Department.

A. SKILLS - RESULTS OF APPLICATION DEVELOPMENT (40%)

B. KNOWLEDGE - SCIENTIFIC WRITING (40%)

C. ATTITUDE & PRESENTATION (20%)

10.2. Grading

The final score is the average of the scores given by the members of Council. The Council is comprised of an instructor and two other members.

11. Learning materials:

Learning materials information	Barcode number
Learning materials (books, textbooks, websites) necessary for students' work will be recommended by instructors.	

Cần Thơ, ngày 27 tháng 09 năm 2022

TL. HIỆU TRƯỞNG
TRƯỞNG KHOA

Nguyễn Hữu Hòa

TRƯỞNG BỘ MÔN

Trương Minh Thái